Arduino Nano Safety Logic Bench Labels

Use these on the switch board: D2-D9 choose +5V for OK or GND for fault. D12 is reset.

Switch / Pin Labels

D2
X1 UL
Upper limit OK

D3
X2 UL
Upper limit OK

D4
Y UL
Upper limit OK

D5
Z UL
Upper limit OK

D6
X1 ALM
Driver alarm OK

D7
X2 ALM
Driver alarm OK

D8
Y ALM
Driver alarm OK

D9
Z ALM
Driver alarm OK

D10
RUN
Output: LOW = run relay on

D11
UL OK
Output: LOW = UL OK relay on

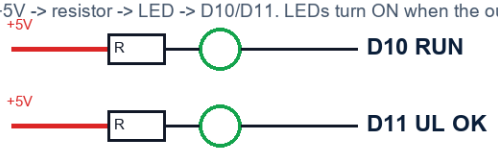
D12
RESET
Input: LOW = pressed

Bench Logic Legend

D2-D9 +5V	OK
D2-D9 GND	FAULT
D12 GND	RESET PRESSED
D10 LOW	RUN OUTPUT ON
D11 LOW	UL OK OUTPUT ON

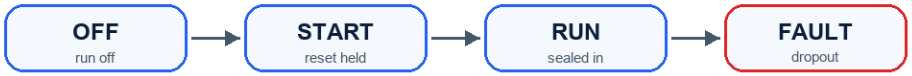
Output LED Check

+5V -> resistor -> LED -> D10/D11. LEDs turn ON when the output pin goes LOW.



Safety Logic Test Scenario Matrix

Set the switches, then compare D10/D11 LEDs. ON means the Arduino output pin is LOW.



#	Scenario	D2-D5 UL	D6-D9 ALM	D12 Reset	D10 RUN	D11 UL OK	What this proves
1	Power-up idle	OK	OK	REL	OFF	ON	Run is off until reset is pressed.
2	Start request	OK	OK	HOLD	ON	ON	Press and hold reset. Run output energizes.
3	Seal-in check	OK	OK	REL	ON	ON	After alarms are OK, release reset; run stays on.
4	Upper limit trip	ONE FLT	OK	REL	OFF	OFF	Any D2-D5 fault drops run and UL OK output.
5	Recover from UL	OK	OK	REL	OFF	ON	Release reset first. Then press reset to restart.
6	Driver alarm trip	OK	ONE FLT	REL	OFF	ON	Any D6-D9 fault drops run but UL OK stays on.
7	Driver startup grace	OK	ONE FLT	HOLD	ON	ON	While reset is held, run may energize before alarms are OK.
8	Release too soon	OK	ONE FLT	REL	OFF	ON	If reset is released before alarms OK, run drops.
9	Finish startup	OK	OK	HOLD->REL	ON	ON	Hold reset until alarms OK, then release.

Fast fault checks: flip any one of D2-D5 to GND -> D10 OFF and D11 OFF. Flip any one of D6-D9 to GND while running -> D10 OFF, D11 stays ON.